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Differentiable Stroke Planning with Dual Parameterization for Efficient and High-Fidelity Painting Creation

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Abstract

In stroke-based rendering, search methods often get trapped in local minima due to discrete stroke placement, while differentiable optimizers lack structural awareness and produce unstructured layouts. To bridge this gap, we propose a dual representation that couples discrete polylines with continuous Bézier control points via a bidirectional mapping mechanism. This enables collaborative optimization: local gradients refine global stroke structures, while content-aware stroke proposals help escape poor local optima. Our representation further supports Gaussian-splatting-inspired initialization, enabling highly parallel stroke optimization across the image. Experiments show that our approach reduces the number of strokes by 30–50%, achieves more structurally coherent layouts, and improves reconstruction quality, while cutting optimization time by 30–40% compared to existing differentiable vectorization methods.

1. Introduction

The pursuit of non-photorealistic rendering, particularly in applications like digital oil painting and interactive sketching, has long been driven by the goal of automatically translating images into expressive, stroke-based artworks. This process, known as stroke-based rendering or image vectorization, fundamentally relies on how strokes are represented and optimized. Existing approaches predominantly fall into two categories, each with significant drawbacks. On one hand, search-based methods [8, 9, 11, 13, 20, 28, 29, 31] trace strokes by following local pixel gradients, often using discrete, rectangular primitives. This leads to structural fragmentation and sensitivity to image noise. Consequently, these methods typically require a vast number of strokes, often in the tens of thousands for a single image to achieve reasonable fidelity, resulting in slow optimization and mediocre reconstruction quality (e.g., PSNR often below 27 dB for complex textures). On the other hand, differentiable optimization methods [18, 22, 32] leverage gradient descent for efficient color fitting. While more stable and faster per stroke, they lack explicit structural priors. The optimization is dominated by pixel-level loss, frequently producing visually disorganized and unstructured stroke layouts that fail to capture the coherent flow of image features, thereby tending to be trapped in local minima and limiting their artistic quality and editability.

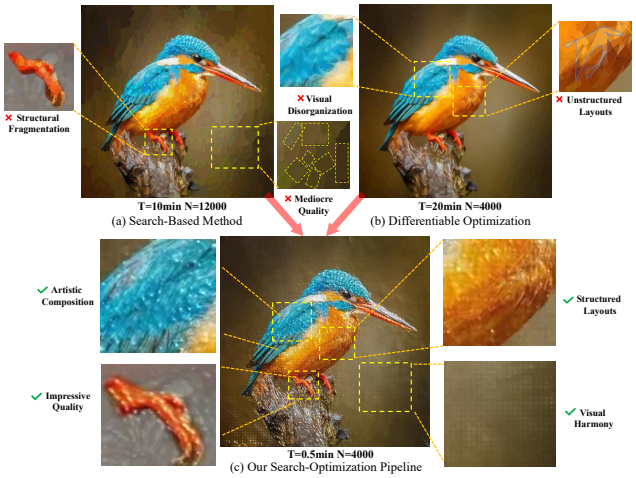


Figure 1. **Comparison with existing stroke-based rendering approaches.** (a) Search-based methods suffer from structural fragmentation and require numerous strokes, leading to slow and time-consuming optimization. (b) Differentiable optimization methods converge faster but often produce visually disorganized and unstructured layouts due to the lack of explicit structural priors. (c) Our method combines discrete structure-aware search with differentiable refinement, producing coherent, visually harmonious, and high-fidelity paintings with significantly fewer strokes.

To overcome these inherent limitations, we introduce a novel hybrid optimization framework that seamlessly integrates discrete structural exploration with continuous gradient-based refinement. At its core is a *dual stroke representation*: a discrete polyline defined by a sequence of anchor points, and a continuous counterpart represented by smooth Bézier control points. These two representations are bidirectionally mapped through a differentiable spline conversion layer, allowing loss gradients to flow from the continuous domain back to influence the discrete structure.

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This coupling enables our two-stage optimization cycle. First, a residual-guided parallel search proactively proposes short, discrete stroke segments in regions with high reconstruction error, effectively capturing major image structures and escaping poor local minima. Second, a differentiable polyline splatting renderer with an anisotropic kernel takes these proposals and refines them using gradient descent, optimizing their color, shape, and transparency for pixel-accurate fitting. A key innovation is our learnable transparency model, which allows strokes to blend smoothly, significantly reducing the number required for high-quality reconstruction. Furthermore, our representation naturally supports a Gaussian-splatting-inspired initialization strategy, where potential stroke seeds are distributed based on image feature density, enabling massive parallel optimization of thousands of strokes from the very beginning. The principal advantages of our system are threefold: 1) it produces structurally coherent and artistically plausible layouts; 2) the resulting vectorized strokes are highly amenable to editing; and 3) it achieves a significant boost in optimization efficiency and final visual fidelity.

Extensive evaluations across multiple benchmarks show that our method significantly outperforms prior work [12, 14, 21, 28, 32] on a wide range of metrics. With 30–50% fewer strokes, our approach achieves higher reconstruction fidelity, highlighting its efficient use of primitive elements; on complex texture datasets, our structure-aware optimization yields a 4–5 dB PSNR gain. Importantly, this improvement comes with substantially reduced runtime—thanks to strategic initialization and a GPU-parallel splatting design, the total optimization time decreases by 30–40%, often converging to high-quality results within minutes. Qualitative comparisons and a user study further confirm that our method produces stroke layouts that are not only more accurate but also visually more coherent and better aligned with salient features such as edges and ridges. Taken together, the combination of efficiency, compact representation, and superior artistic quality marks a meaningful step toward practical, high-quality stroke-based rendering.

2. Related Work

2.1. Stroke Planning

Stroke planning focuses on organizing brush strokes to effectively represent image structures and styles. Early search-based methods [9, 11, 20, 29] trace strokes by following local gradients. Subsequent search-based techniques introduced image segmentation and adaptive sampling to improve reconstruction quality [8, 13, 28, 31]. However, these methods often produce structurally fragmented results and require a large number of strokes with slow optimization. Additionally, they remain prone to local minima due to discrete stroke placement. Recently, neural networks like RNNs [10], Transformers [21, 24, 26], and RL-based approaches [12, 14, 15, 19, 25] have been adopted for stroke sequence generation, but they face challenges in both generalization and controllability. Meanwhile, differentiable rendering methods [3–5, 18, 22, 32] employ gradient-based optimization for efficient color fitting, enabling faster convergence per stroke. However, they often produce visually disorganized layouts due to a lack of structural awareness. Our approach addresses these limitations through a dual representation that combines discrete polylines with continuous Bézier control points.

2.2. Stroke Representation

Stroke representation is fundamental to stylized brushstroke rendering, as it impacts the visual richness and style fidelity of the output. Early methods such as [20, 23] model strokes using rectangular segments, which are simple to render but often fail to capture smooth contours, leading to fragmented results and an excessive number of short strokes. To better approximate natural brush trajectories, curved representations have been widely adopted, including B-splines [11] and Bézier curves [8, 17, 18, 27]. While these support continuous paths and richer shapes, they often suffer from slow rendering and local redundancies, and still lack higher-level structural organization. Other studies enrich stroke appearance using texture templates collected from real artworks [28, 31], though such methods frequently exhibit texture discontinuities and limited adaptability. Physics-based modeling techniques simulate pigment-canvas interactions to enhance realism, such as viscous fluid deposition [2, 7] and particle-based pigment dynamics [6]. These achieve compelling material effects, but rely on complex simulation and are difficult to optimize at the stroke level.

3. Methodology

Our method is motivated by a key observation: the complementary strengths of discrete search and continuous optimization can be harnessed through a **dual stroke representation** that enables them to work in concert. This synergy enables efficient exploration of large-scale structures while achieving precise, pixel-level fidelity. Given a target image $I_{\text{gt}} \in \mathbb{R}^{H\times W\times 3}$, our goal is to find a set of strokes $\mathcal{S} = \{s_i\}$ whose rendering I minimizes the reconstruction loss $\mathcal{L}(I, I_{\text{gt}})$. Each stroke s_i is primarily defined by its spatial trajectory and appearance.

The core of our framework is an iterative **search-and-optimize** pipeline, built upon our dual stroke representation. As illustrated in Fig. 2, each iteration consists of two synergistic stages: 1) **Structure-Aware Search**: Proposes new stroke structures in under-reconstructed regions; 2) **Gradient-Guided Optimization**: Refines the parameters of all existing strokes. These steps are followed by a **Differentiable Polyline Renderer** that efficiently com-

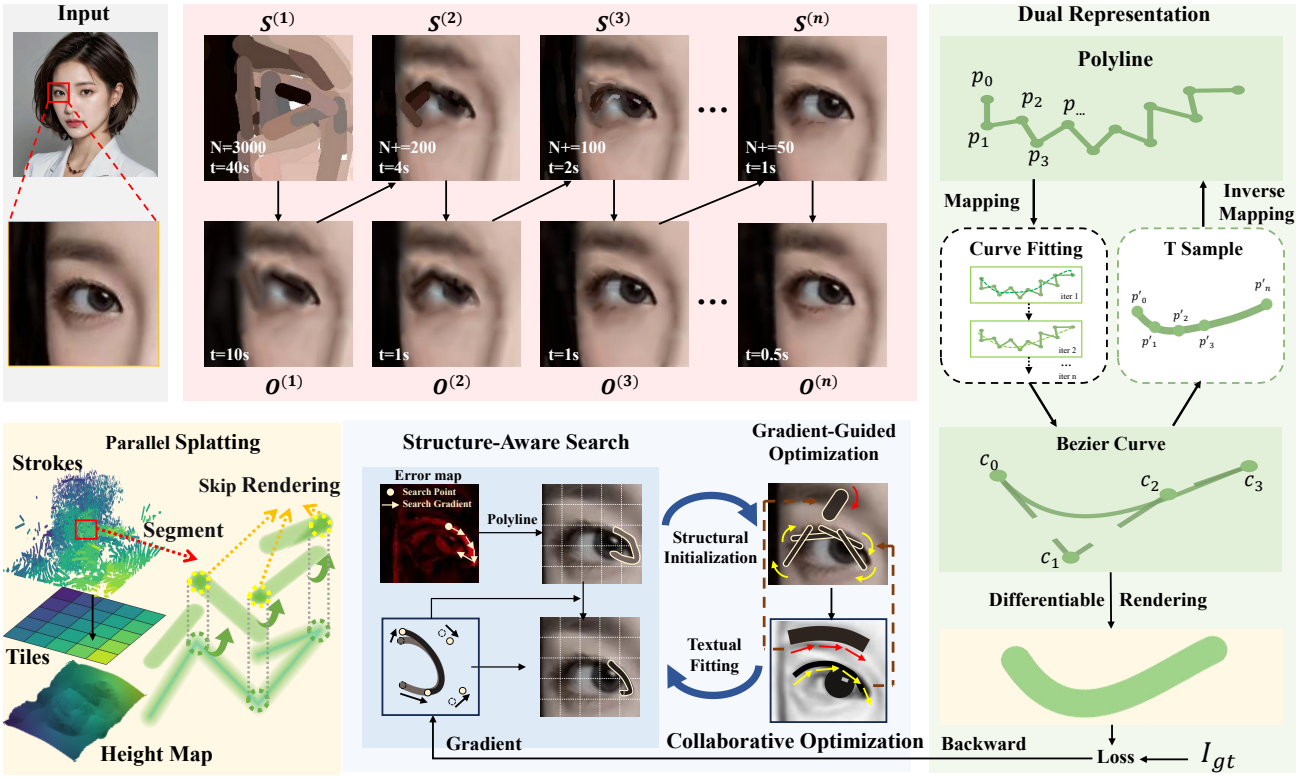


Figure 2. **Framework.** Given a target image I_{gt}, our method iteratively reconstructs it through a dual-stage *search-and-optimize* pipeline. At iteration i, the structure-aware search step $S(i)$ samples seeds from high-residual regions using NMS and traces geometry-aligned polylines along image gradients. These polylines are then mapped, via a bidirectional differentiable dual representation, to Bézier curves and refined in the optimization step $O(i)$ using gradients from our differentiable polyline splatting renderer. All strokes are splatted in parallel to form the canvas, and the process rapidly converges, with most structure recovered in the first iteration.

posites strokes onto the canvas, along with coherent stroke height regression and lighting. The pipeline repeats until convergence, progressively building a coherent and compact set of strokes that accurately approximates the target image.

3.1. Dual Stroke Representation

We introduce a dual representation to bridge the gap between discrete search and continuous gradient optimization, enabling them to collaboratively refine the same stroke to pursue both structural and fine-grained stroke fitting harmony. To this end, a stroke s_i is simultaneously represented in both forms: 1) **Discrete Polyline**: A sequence of ordered vertices, $\mathbf{P}_i = [\mathbf{p}_{i,0}, \mathbf{p}_{i,1}, \ldots, \mathbf{p}_{i,M_i-1}]$, which is intuitive for structural proposal and manipulation. 2) **Continuous Bézier Curve**: A smooth parametric curve defined by a set of control points $\mathbf{C}_i = [\mathbf{c}_{i,0}, \mathbf{c}_{i,1}, \ldots, \mathbf{c}_{i,K-1}]$, which is ideal for differentiable refinement.

These two representations are bidirectionally mapped through a shared parameterization, forming the backbone of our cooperative optimization framework.

Mapping: Polyline to Bézier (Differentiable Fitting) To convert the discrete polyline $\mathbf{P}_i$ into a continuous Bézier curve, we first assign a normalized parameter $t_{i,m} \in [0, 1]$ to each vertex $\mathbf{p}_{i,m}$ using chord-length parameterization along the polyline. The goal is to find the Bézier control points $\mathbf{C}_i$ that best approximate the polyline vertices. The position of a point on the Bézier curve at parameter t is given by:

$$\mathbf{B}(t; \mathbf{C}_i) = \sum_{k=0}^{K-1} \mathbf{c}_{i,k} \cdot B_{k,K-1}(t), \tag{1}$$

where $B_{k,K-1}(t)$ is the Bernstein basis polynomial of degree $K-1$. We solve for the optimal control points $\mathbf{C}_i^*$ by minimizing the least-squares error between the original polyline and the fitted curve:

$$\mathbf{C}_i^* = \arg\min_{\mathbf{C}_i} \sum_{m=0}^{M_i-1} \|\mathbf{p}_{i,m} - \mathbf{B}(t_{i,m}; \mathbf{C}_i)\|_2^2 . \tag{2}$$

This convex problem has a closed-form solution via the pseudoinverse, allowing the fitting process to be trivially differentiated, thus enabling gradient flow from the continuous curve back to the polyline vertices.

Mapping: Bézier to Polyline (Differentiable Sampling) The reverse mapping is straightforward and fully differentiable. Given a set of Bézier control points $\mathbf{C}_i$, we can sample the corresponding discrete polyline $\mathbf{P}_i$ by evaluating the Bézier curve at a fixed set of parameters $\{t_{i,m}\}$:

$$\mathbf{p}_{i,m} = \mathbf{B}(t_{i,m}; \mathbf{C}_i) \tag{3}$$

This sampling operation is a linear combination of the control points and thus differentiable with respect to $\mathbf{C}_i$.

This bidirectional, differentiable bridge allows gradient signals from the continuous optimization to refine the discrete structure proposed by the search module, ensuring that both global layout and local details are optimized.

3.2. Structure-Aware Stroke Search

This module is responsible for proactively discovering strokes that capture prominent image structures. The core of our search is a **gradient-driven tracing** mechanism, which directly leverages the continuous gradient field to guide the discrete growth of polylines, ensuring that the proposed structures are both geometrically salient and optimized for loss reduction from their inception.

The search begins with the reconstruction residual R^t and its gradient field $\mathbf{G}^t$ at iteration t:

$$R^t(\mathbf{x}) = \|I^t(\mathbf{x}) - I_{\text{gt}}(\mathbf{x})\|_2^2, \quad \mathbf{G}^t(\mathbf{x}) = \nabla R^t(\mathbf{x}). \tag{4}$$

Seed points $\{\mathbf{x}_j\}$ are sampled from the distribution of R^t using non-maximum suppression. For each seed, rather than merely initializing a fixed-direction segment, we **trace a polyline** by iteratively following the path that minimizes the reconstruction error.

Starting from $\mathbf{p}_0 = \mathbf{x}_j$, we grow the polyline by repeatedly stepping along a gradient-flow direction $\mathbf{d}_k$:

$$\mathbf{p}_{k+1} = \mathbf{p}_k + \eta \cdot \mathbf{d}_k, \tag{5}$$

where η is an adaptive step size. Intuitively, $\mathbf{d}_k$ performs a **gradient ascent** on the residual field, pushing the stroke to cover high-error regions.

To prevent the trace from being trapped by local noise, the direction $\mathbf{d}_k$ at each step is a weighted combination of the local gradient and the previous direction $\mathbf{d}_{k-1}$, encouraging smoothness:

$$\tilde{\mathbf{d}}_k = \frac{\mathbf{G}^t(\mathbf{p}_k)}{\|\mathbf{G}^t(\mathbf{p}_k)\| + \epsilon}, \quad \mathbf{d}_k = \lambda \cdot \tilde{\mathbf{d}}_k + (1-\lambda) \cdot \mathbf{d}_{k-1}. \tag{6}$$

The tracing terminates when the current stroke no longer reduces the reconstruction loss, and the overall search exceeds a threshold of consecutive failures.

This method produces a set of discrete polylines $S_{\text{search}} = \{\mathbf{P}_i\}$ where each vertex is *explicitly dictated by the gradient flow*. The resulting strokes are therefore not only structurally coherent but also primed for loss reduction. They are then directly passed through our dual representation to become Bézier curves, ready for fine-grained, differentiable adjustment in the next stage. In this way, the gradient field serves as a continuous guide throughout the entire discrete search process, achieving true collaboration between the two optimization paradigms.

3.3. Gradient-Guided Stroke Optimization

The stroke set $S_{\text{search}} = \{\mathbf{P}_i\}$ proposed by the discrete search module provides a structurally coherent initialization in the form of discrete polylines. This module refines the set within the **continuous Bézier representation** $\{\mathbf{C}_i\}$ to achieve pixel-level fidelity. The key to our efficient co-design is that the search module has already solved the large-scale structural assignment, allowing the gradient-based optimizer to focus on fine-grained adjustments without the risk of collapsing into globally poor layouts.

All strokes are optimized concurrently. The Bézier control points $\mathbf{C}_i$ for each stroke are uniformly sampled to produce a dense, differentiable polyline $\tilde{\mathbf{P}}_i$ for rendering. The entire set is then rendered in a single forward pass using our differentiable splatting kernel (Sec. 3.4) to produce the canvas I_{render}. We minimize the reconstruction objective

$$\mathcal{L} = \|I_{\text{render}} - I_{\text{gt}}\|_2 + \lambda_\ell \mathcal{L}_{\text{len}} + \lambda_w \mathcal{L}_{\text{width}}, \tag{7}$$

where $\mathcal{L}_{\text{len}}$ and $\mathcal{L}_{\text{width}}$ regularize stroke length and width. Gradients $\nabla_{\mathbf{C}_i, \mathbf{c}_i, \mathbf{o}_i, w_i} \mathcal{L}$ are fully backpropagated through the renderer, enabling end-to-end refinement of geometry, color, opacity, and width.

We employ an adaptive optimization strategy similar to 3DGS [16] for stable convergence. Since our strokes are typically long and structurally important, directly splitting or pruning them would damage global contour coherence. We therefore only reinitialize strokes with near-zero opacity and minimal reconstruction loss, letting the structure-aware search module assign their new parameters. This preserves structural integrity while ensuring stable and efficient optimization.

This continuous refinement directly benefits from the structural prior provided by the search stage. Since the initial strokes are already well-placed along salient edges and ridges, the gradient descent efficiently fine-tunes their geometry and appearance rather than attempting costly global reconfiguration. This division of labor—**coarse exploration via search, fine exploitation via gradients**—is the cornerstone of our efficiency.

Conversely, the gradient optimization legitimizes and enhances the proposals from the discrete search. A search-proposed stroke might be geometrically approximate; the gradient refinement **corrects** its precise placement, width, and color, transforming a structurally correct but coarse proposal into a visually accurate and seamless element of

the final painting. This synergistic cycle iterates until convergence, progressively building a compact and coherent stroke set S^*.

3.4. Differentiable Polyline Renderer

To enable end-to-end optimization while maintaining high rendering efficiency, we design a differentiable renderer that approximates each stroke segment as a soft, anisotropic splat. This offers a favorable trade-off between the computational cost of exact Bézier rasterization and the need for full differentiability.

Each stroke, after being sampled from its Bézier representation into a polyline $\tilde{\mathbf{P}}_i$, is decomposed into consecutive segments. The influence of a segment $e_{i,j}$ on a pixel $\mathbf{x}$ is modeled by an anisotropic kernel based on the shortest distance $d_{i,j}(\mathbf{x})$:

$$k_{i,j}(\mathbf{x}) = \frac{\sigma\left(\frac{w_i/2 - d_{i,j}(\mathbf{x})}{\tau}\right) - \sigma\left(\frac{-w_i/2}{\tau}\right)}{1 - 2 \cdot \sigma\left(\frac{-w_i/2}{\tau}\right)}, \tag{8}$$

where w_i is the stroke width, τ a softness parameter, and σ the logistic function. The resulting opacity is $\alpha_{i,j}(\mathbf{x}) = \mathbf{o}_i \cdot \max(0, k_{i,j}(\mathbf{x}))$.

All segments are composited onto the canvas in a single, GPU-parallel pass using front-to-back alpha blending. Let $\pi(\mathbf{x})$ denote the sequence of stroke segments overlapping pixel $\mathbf{x}$, sorted from front to back:

$$C(\mathbf{x}) = \sum_{(i,j)\in\pi(\mathbf{x})} \alpha_{i,j}(\mathbf{x})\, \mathbf{c}_i \prod_{(p,q)<(i,j)} (1 - \alpha_{p,q}(\mathbf{x})). \tag{9}$$

A key to our efficiency is that this splatting paradigm, inspired by recent advances in Gaussian splatting, enables highly parallelized rasterization of all stroke segments. This directly enables the **Gaussian-splatting-inspired initialization** in Sec. 3.3, where a large population of stroke primitives can be placed and optimized simultaneously from the first iteration, dramatically accelerating convergence.

The entire rendering process is fully differentiable, allowing gradients to flow from the total loss $\mathcal{L}$ back to the stroke parameters $\mathbf{C}_i$, $\mathbf{c}_i$, $\mathbf{o}_i$, and w_i. This efficient, parallel, and differentiable renderer is the backbone that makes our iterative search-and-optimize pipeline practical.

To further enhance the physical realism of the painting, we regress a height value h_i for each stroke, enabling advanced shading effects via relighting. The initial height is estimated from the target image using a pre-trained monocular depth estimation model (e.g., Depth Anything [30]), which provides a dense geometry prior. This initial prediction is then refined with a continuity constraint, encouraging smooth height transitions between spatially adjacent strokes. The final height field allows us to render the painting under novel lighting conditions, simulating the tangible surface relief characteristic of impasto oil paintings.

4. Experiments

4.1. Experimental Setups

Implementation details. All experiments are implemented in PyTorch and run on a single NVIDIA RTX 4090 GPU. Since our pipeline already achieves strong convergence in the first iteration, we use a default of three search–optimization iterations. During the **search phase**, we extract seeds from the top 12% residual pixels using a 7×7 NMS window. We trace strokes using an adaptive step, starting at 1.2 pixels with a smoothing factor of $\lambda = 0.8$. Strokes grow up to 20 vertices dynamically and are accepted only if they reduce the reconstruction loss by at least 0.01. The local search halts after 20 consecutive rejections.

During the **optimization phase**, discovered polylines are converted into piecewise cubic Bézier curves and sampled at 10 discrete points for differentiable rendering. We jointly optimize all stroke parameters using Adam for up to 4000 iterations per stage or until convergence. The base learning rate is 0.01, with color and width updates scaled by 0.1 and 0.01, respectively.

Dataset & Evaluation. We evaluate on two complementary datasets. (1) The DIV2K validation set [1], a high-resolution and diverse natural-image benchmark resized to 1200×1200 for consistent stroke resolution. (2) The Im2Oil gallery dataset [28], featuring classical oil paintings at 600 × 800 with characteristic textures and styles. We report PSNR, SSIM, LPIPS, stroke count, per-iteration runtime, and a perceptual user study.

4.2. Comparative Analysis with SOTA

We compare our approach with five state-of-the-art stroke-based rendering methods, including Im2Oil [28], Compositional Neural Painting (CNP) [12], Paint Transformer [21], Learning to Paint [14], and Stylized Neural Painting (SNP) [32]. We employ 16K strokes for the DIV2K dataset and 8K strokes for the Im2Oil dataset based on image resolution, to align parameters across all baselines.

Qualitative comparisons in Fig. 3 show that our method consistently produces clearer structural boundaries, cleaner textures, and more coherent stroke flows on artistic images. By leveraging our Dual Stroke Representation and joint search-optimization pipeline, we effectively preserve thin structures and smooth regions, whereas baselines struggle with noisy, overlapping strokes.

Quantitatively, Table 1 shows our superior reconstruction accuracy across both datasets. Notably, the substantial improvements in SSIM highlight our method's capability in preserving global structures and semantic coherence. Beyond fidelity, our method exhibits remarkable computational efficiency. By synergizing the dual stroke representation with GPU-parallel differentiable splatting, our pipeline substantially accelerates the optimization process, running

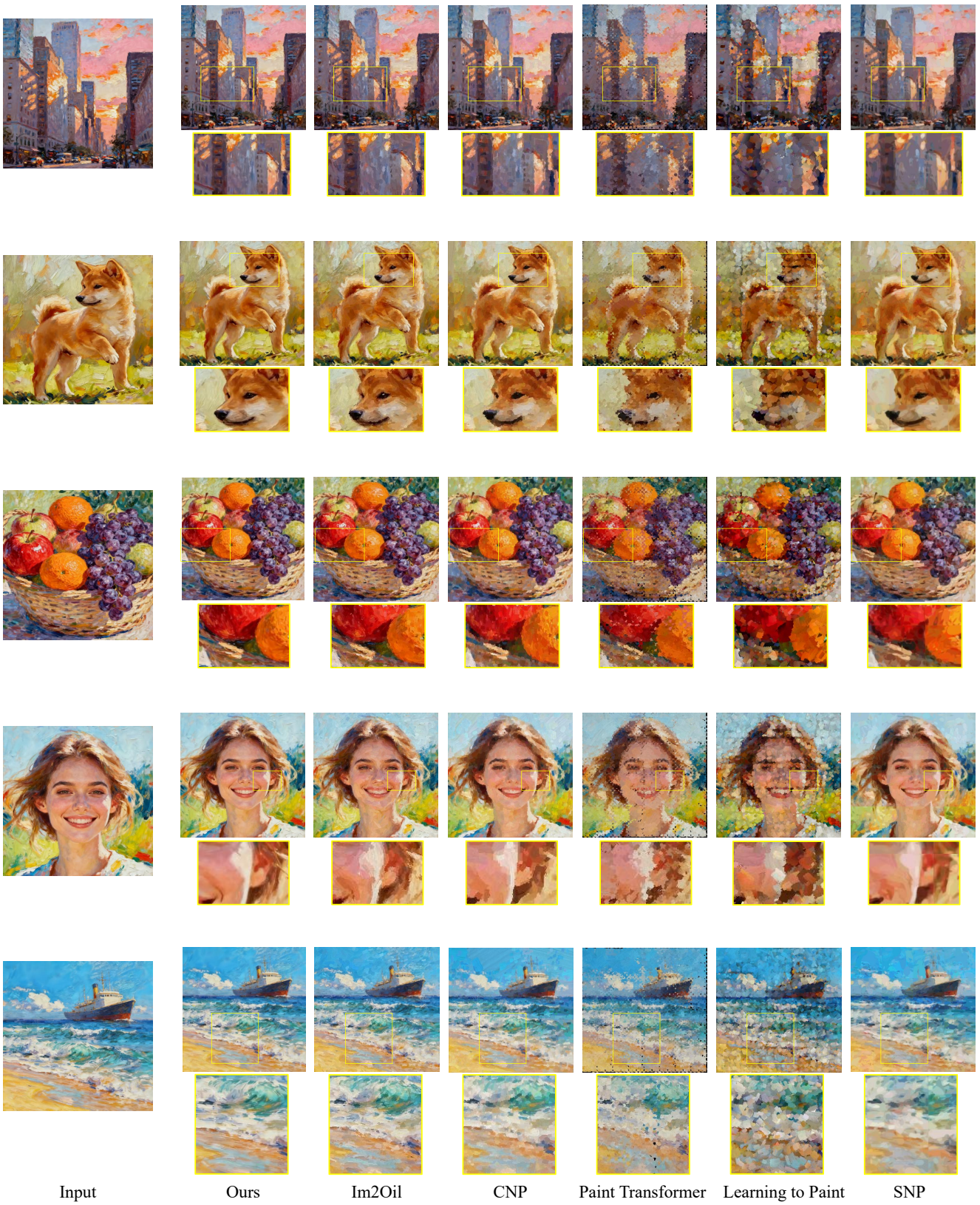


Figure 3. **Qualitative comparison with state-of-the-art methods.** Our method produces clearer contours, cleaner textures, and more coherent stroke flows across both natural and synthetic domains. In contrast, baseline approaches often generate noisy strokes or lose structural consistency, particularly under limited stroke budgets.

Method	PSNR↑	SSIM↑	LPIPS↓	Time (s)↓
DIV2K Validation				
Im2Oil [28]	27.59	0.72	0.211	727.8
CNP [12]	27.91	0.64	0.296	125.5
Paint Transformer [21]	23.83	0.63	0.378	414.1
Learning to Paint [14]	27.19	0.70	0.330	167.9
SNP [32]	20.63	0.42	0.405	5837.2
Ours	**32.16**	**0.93**	**0.076**	**87.6**
Gallery Dataset				
Im2Oil [28]	28.54	0.71	0.204	392.8
CNP [12]	28.46	0.68	0.288	61.3
Paint Transformer [21]	24.46	0.61	0.395	197.4
Learning to Paint [14]	28.58	0.73	0.318	80.6
SNP [32]	20.87	0.49	0.433	2868.1
Ours	**32.53**	**0.86**	**0.192**	**42.7**

Table 1. Quantitative comparison with SOTA SBR methods.

Method	Structure↑	Texture↑	Color↑	Overall↑
Im2Oil [28]	3.42	3.28	3.55	3.38
CNP [12]	3.65	3.12	3.08	3.31
Paint Transformer [21]	2.98	3.05	2.87	2.97
Learning to Paint [14]	3.51	3.42	3.47	3.47
SNP [32]	2.45	2.38	2.31	2.41
Ours	**4.32**	**4.28**	**4.41**	**4.35**

Table 2. User study results (1–5 Likert ratings; higher is better). Our method achieves the best rank across all perceptual criteria.

significantly faster than existing SBR baselines.

To assess perceptual quality beyond pixel-level metrics, we conducted a user study with 100 participants, including 60 from a university campus and 40 from an online platform; about 30% had formal training in visual arts or design. In each trial, participants were shown a set of source images together with six corresponding oil paintings generated by our method and five baselines, displayed in randomized order. They were asked to select the best painting and then marked the regions they found most compelling by drawing bounding boxes, encouraging fine-grained judgments. Participants further rated each painting across four perceptual criteria commonly used in non-photorealistic rendering: *Structural Fidelity*, *Textural Realism*, *Color Accuracy*, and *Overall Preference*, each on a 1–5 Likert scale. Table 2 shows that our method ranks highest across all criteria, confirming its perceptual superiority.

4.3. Tunable Aesthetics and Physical Realism

Beyond reconstruction fidelity, our pipeline offers highly tunable aesthetics and physical realism. Stroke appearance is flexibly controlled via the hardness parameter τ: higher values ($\tau = 0.7$, evaluated in Sec. 4.2) yield smooth, blended photorealism, whereas lower values ($\tau = 0.1$) produce sharp, distinct brush-like strokes that vividly reveal the structural layout. Under this hard-edge setting, Fig. 4

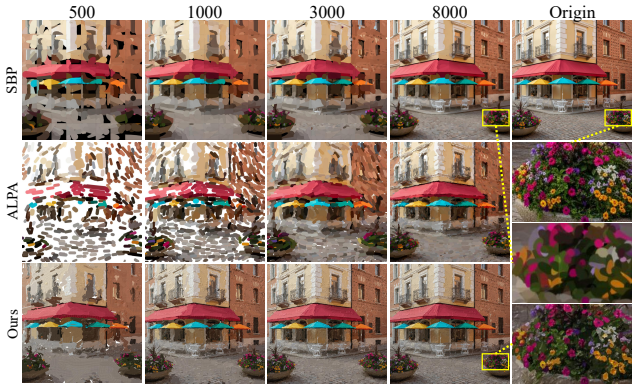


Figure 4. Progressive painting process with height-field relighting.

compares our progressive rendering against ALPA [13] and SBP [8], two recent semantic SBR baselines. Beyond achieving superior fidelity, structural coherence, and stroke efficiency, our method transcends standard 2D image generation. By rendering splatting-based height fields with physics-based lighting, we produce a volumetric 3D model with tangible surface relief, maintaining expressive stroke flows and strong structural coherence even in this physically realistic, hard-edge regime.

4.4. Ablation Study

4.4.1. Analysis of Iterative Optimization Dynamics

Unless otherwise specified, all ablation experiments are conducted on the DIV2K dataset. The core of our method lies in the *Iterative Optimization Dynamics*, for which we set the number of iterations to $t = 3$ in our default configuration. In this section, we conduct ablation experiments to evaluate both the effectiveness and efficiency of the iterative process. As shown in Table 3, we compare the painting reconstruction quality and runtime cost for $t = 1$ through $t = 4$. When $t = 1$, our method already achieves highly competitive reconstruction performance in under 50 seconds. Increasing the number of iterations further improves reconstruction quality with only a marginal increase in optimization time, and we find $t = 3$ to offer the best trade-off between performance and efficiency.

The qualitative results in Figure. 5 further validate these observations. With just a single iteration, our approach already produces stroke-based renderings with strong artistic style and visual realism. As the number of iterations increases, the generated paintings progressively refine geometric precision, stroke smoothness, and color consistency.

4.4.2. Effectiveness of the Dual Stroke Representation

We evaluate the impact of our dual stroke representation by comparing four variants: 1) *Search-only*, where strokes are produced by the structure-aware search but not refined; 2) *Polyline-only optimization*, which optimizes polyline vertices without conversion to Bézier curves; 3) *Bézier-only*

Iter.	Stage	PSNR↑	SSIM↑	LPIPS↓	Time (s)
$t=1$	Search	20.3	0.48	0.323	35.6
	Optimize	28.4	0.82	0.214	47.2
$t=2$	Search	29.6	0.83	0.215	59.0
	Optimize	30.4	0.85	0.194	65.5
$t=3$	Search	29.9	0.84	0.197	75.6
	Optimize	32.2	0.93	0.076	87.6
$t=4$	Search	31.6	0.89	0.112	89.6
	Optimize	32.4	0.93	0.073	95.2

Table 3. Iteration-by-iteration analysis.

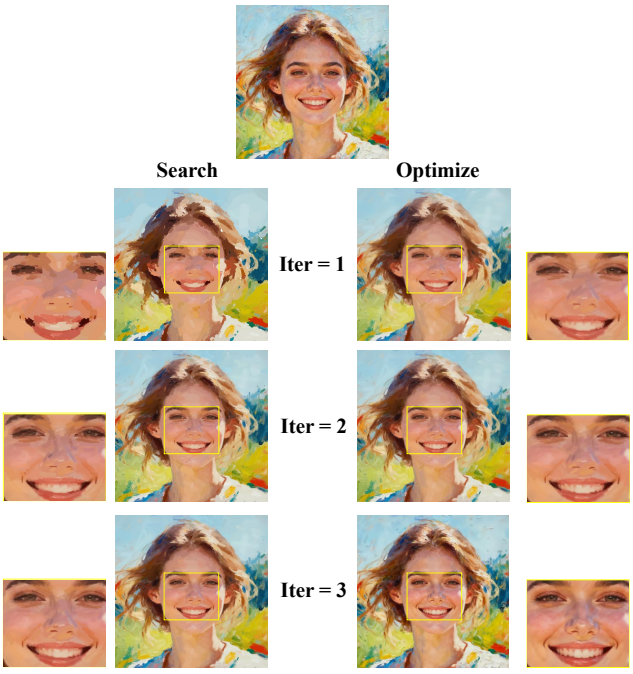


Figure 5. Qualitative visualization of intermediate canvases at iterations $t = \{1, 2, 3\}$. Iteration 1 establishes coarse structural coverage via search, while later iterations progressively refine geometric precision, stroke smoothness, and color consistency.

optimization, where Bézier refinement is performed without discrete search initialization; 4) *Full model*, our joint search–optimization pipeline using the dual representation.

Due to the prohibitive rendering cost of pure Bézier optimization, we limit the Bézier-only variant to 2,000 strokes, while other variants use 16,000 strokes. As shown in Table 4, search-only reconstruction captures coarse structure but lacks fine detail; polyline-only optimization suffers from unstable gradients; Bézier-only optimization improves smoothness but often misses global structure due to the absence of discrete guidance. In contrast, our full model consistently achieves the highest fidelity and the lowest perceptual error, demonstrating that discrete search provides strong structural priors while Bézier curves offer a well-behaved space for continuous refinement. These results confirm that the dual representation is essential for both accuracy and stability.

Variant	PSNR↑	SSIM↑	LPIPS↓	Time (s)↓
Search-only	27.8	0.76	0.227	57.2
Polyline-only optimization	31.2	0.89	0.126	**31.6**
Bézier-only optimization	30.4	0.86	0.138	3978.4
Full model (ours)	**32.16**	**0.93**	**0.076**	87.6

Table 4. Ablation on the dual stroke representation.

Variant	PSNR ↑	SSIM ↑	LPIPS ↓
Search components			
w/o residual seeding	26.6	0.68	0.263
w/o gradient tracing	26.3	0.71	0.249
Full search	**27.8**	**0.76**	**0.227**
Optimization components			
w/o opacity reinit	30.3	0.85	0.158
w/o loss reinit	29.9	0.86	0.152
w/o any reinit	28.1	0.83	0.173
Full optimization	**31.2**	**0.89**	**0.126**

Table 5. Ablation study on search and optimization components.

4.4.3. Ablation on Search and Optimization Components

As detailed in Table 5, we ablate key components of our pipeline. For the search module, removing residual-guided seeding fails to target salient regions, while disabling gradient-driven tracing yields fragmented stroke geometry. Both are essential for structurally meaningful proposals. For the optimization module, disabling opacity reinitialization retains visually negligible strokes, hindering convergence, while removing loss reinitialization fails to correct misaligned geometry. Eliminating both yields the worst performance due to persistently collapsed strokes. Conversely, our adaptive approach selectively resets underperforming strokes, yielding stable refinement and yields optimal fidelity. Ultimately, the search stage provides strong structural priors, and adaptive reinitialization ensures stable gradient-based refinement.

5. Conclusion

We couple discrete polylines with continuous Bézier curves via a differentiable bidirectional mapping, allowing discrete search to establish global structure while gradients refine local details. Integrated with our splatting-based renderer, this approach enables fast optimization and yields high-fidelity painterly renderings using height-based shading.

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